Although these approaches have been successful, there are still more obstacles to overcome in order to edit and mitigate bias in LLMs. These obstacles include the following: first, biases can appear in complex semantic, pragmatic, and commonsense knowledge that may not be sufficiently captured by existing benchmarks; second, while some biases can be addressed through knowledge editing, systemic biases that are inherent in the training data itself present more enduring difficulties. Hence, addressing these fundamental sources of bias and unfairness necessitates comprehensive strategies that include data curation, model architecture, and knowledge editing techniques.

Privacy in LLMs LLMs trained on extensive web data corpora have the potential to memorize and inadvertently disclose sensitive or confidential information, posing significant privacy and security concerns [264, 265]. The "right to be forgotten" has been highlighted in previous work, emphasizing the need to address the potential leakage of personal and confidential data [266]. Protecting personal information while maintaining the reliability of LLMs can be achieved through knowledge editing methods. For instance, Jang et al. [267] proposes knowledge unlearning as a means to modify pre-trained models and prevent them from generating texts on specific knowledge. Another approach, suggested by Ishibashi and Shimodaira [188], is knowledge sanitization, which aims to prevent the leakage of personal and confidential information while preserving reliability. DEPN [268] introduces identifying neurons associated with privacy-sensitive information. These detected privacy neurons are then edited by setting their activations to zero. Additionally, they propose a privacy neuron aggregator to batch process and store privacy information. Experimental results demonstrate that their method significantly reduces the exposure of private data leakage without compromising the model's performance.

In the context of multi-modal models, Chen et al. [269] proposes the PrivQA dataset for protecting personal information. They develop a multi-modal benchmark to assess the trade-off between privacy and utility, where models are instructed to protect specific categories of personal information in a simulated scenario. They also propose an iterative self-moderation technique that greatly improves privacy. Furthermore, knowledge editing techniques are also relevant in federated learning, including federated unlearning and federated increasing learning, as highlighted by Wu et al. [270]. Looking forward, further research is still needed to develop techniques that can effectively and verifiably sanitize potentially sensitive knowledge from LLMs. Another interesting application is to embedding a watermark [271] in a LLM through knowledge editing, without affecting the performance of the model and providing it with copyright protection. Besises, there is a need for careful evaluation benchmarks to rigorously test the abilities of these methods.

6.4 Human-Computer Interaction: Personalized Agents

Millions of years of evolution have enabled humans to achieve intelligence through genes and learned experiences. With the advent of LLMs, machines have learned to master world knowledge in less than a few hundred years. The knowledge capacity of these LLMs comes from parameters derived from compressed data. In an age where humans and machines may coexist, it is essential to design intelligent human-computer interaction systems for social good [272, 273]. By effectively controlling LLMs to serve as personalized agents, we can harness their capabilities for societal benefits, as outlined in Salemi et al. [274]. Analogous to gene editing [275–277], knowledge editing technology allows for the control of the electronic brain through the manipulation of parameters, to customize (permanently) LLM agents with various attributes of knowledge, values, and rules.

Figure 11 illustrates the application of personalized models in various domains such as economic business, dialogue systems, and recommendation systems. Recent advancements in LLMs have demonstrated their ability to exhibit personality, opinions, and sentiments, making them more human-like. This has sparked a growing interest in developing personalized LLMs. Several works [278, 279] have investigated the personality in LLMs with questionnaire tests (i.e. MBTI) and other psychological theories. Tu et al. [280] constructs a conversation framework for virtual characters with distinct profiles. Mao et al. [281] proposes a new knowledge editing task to edit LLM's personality. Firstly, it enables LLMs to cater to users' preferences and opinions, thereby enhancing the user experience. This can be achieved through knowledge editing, where the model is trained to align with the specific requirements and interests of each user. An emotion benchmark [282] is also proposed to measure LLM's emotion.

Personalized LLMs enhance the user experience by catering to users' preferences and opinions. Knowledge editing is a key technique in achieving this. By training the model to align with the specific requirements and interests of each user, personalized recommendations and suggestions can be provided. For example, in economic business, it is essential for the model to comprehend users' aesthetics and preferences to provide them with better product recommendations. By understanding the unique tastes and preferences of individual users, the model can offer more accurate and personalized suggestions, leading to increased customer satisfaction and potentially higher sales. Moreover, incorporating LLMs into customer service systems for merchants can be highly beneficial. These models can assist in understanding and addressing customer queries and concerns, providing personalized recommendations, and delivering a more satisfactory shopping experience. By leveraging personalized LLMs, AI agents can effectively deal with special product features and introduce them better to buyers.

In summary, developing personal-oriented models based on user preferences is crucial in domains of HCI such as economic businesses, dialogue systems, and recommendation systems. Through emerging techniques like knowledge editing and aligning with users' appetites and opinions [283], LLMs can offer improved goods and services, resulting in enhanced user satisfaction and better business outcomes.

7 Discussion and Conclusion

In this study, we highlight the challenges inherent to present-day knowledge editing and introduce a new benchmark for diverse editing tasks. While current methods have shown efficacy in certain areas, **significant issues** remains for enhancement:

- The current language model architecture of Transformers is fundamentally based on the next token prediction task, yet the underlying mechanism remains opaque. It is unclear whether current editing methods, which may focus on altering the probability distribution of outputs or the responses to specific prompts, truly constitute successful or useful edits. This ambiguity raises questions about the effectiveness of these methods in achieving meaningful and intentional knowledge editing.

- Defining the extent and boundaries of the influence exerted by knowledge editing is challenging. Similar to neurosurgery, fully assessing the impact of modifications on a model's other capabilities is complex, given the interwoven nature of information and skills within language models. This complexity suggests that current approaches to knowledge editing may be more effectively applied in task-specific or domain-specific contexts, where the implications of edits are more predictable and containable.

- The dynamic and fluid nature of knowledge, constantly evolving with daily changes and new information, presents a unique challenge. Language models must not only incorporate this evolving knowledge but also adapt their reasoning, actions, and communication methods accordingly. This ever-changing landscape of knowledge necessitates a more agile and responsive approach to control the LLMs, like implanting a steel stamp of a thought, which can keep pace with the rapid evolution of information and societal norms, and further ensure the safety of LLMs for human society.

However, just as Pinter and Elhadad [184] argues, the stochastic nature of LLMs is not only a source of complexity but also a wellspring of creativity and adaptability in various scenarios. Hence, the potential of knowledge editing is still worth exploring. Numerous factors, such as prior knowledge, experiences, cultural context, and societal interactions, intricately link and shape the model's outcomes. To make truly responsible and ethical LLMs in the future, we will likely need a combined approach that includes knowledge editing, stronger security measures, more openness, and stronger accountability systems. Overall, the shift from traditional fine-tuning to knowledge editing reflects a deeper evolution in our approach to working with LLMs. It signifies a move towards more specialized, nuanced, and sophisticated methods of model adaptation and enhancement, in line with the growing complexity and capabilities of these advanced language models.

Broader Impacts

Knowledge editing, in the context of LLMs, refers to methodologies and techniques aimed at updating and refining these models more efficiently. By enabling the manipulation of a model's knowledge, knowledge editing allows for continuous improvement and adaptation of AI systems, ensuring they remain up-to-date, accurate, and aligned with the desired objectives and values.

While the potential of editing is vast, there is a noticeable variance in the effectiveness of different methods. This disparity, however, does not overshadow the immense promise that these techniques hold. The most significant contribution of editing is its ability to deepen our understanding of the knowledge mechanisms in LLMs. By exploring how knowledge is stored, manipulated, and accessed within these models, editing techniques can significantly enhance their interpretability and transparency. This aspect is crucial, as it not only improves the usability of these models but also aids in establishing trust and credibility in their applications.

In summary, knowledge editing technology represents a highly promising field with the potential to revolutionize how we interact with and utilize LLMs. Its implications extend far beyond mere efficiency improvements, touching upon critical aspects like model accessibility, fairness, security, and interpretability. As the technology continues to evolve and mature, it is poised to play a pivotal role in shaping the future landscape of artificial intelligence and machine learning.

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Open Resources

KnowEdit (Huggingface): https://huggingface.co/datasets/zjunlp/KnowEdit.

EasyEdit (Github): https://github.com/zjunlp/EasyEdit.

Contributions

The contributions of all authors are listed as follows: Ningyu Zhang, Yunzhi Yao, Peng Wang, Bozhong Tian and Shumin Deng initiated and organized the research. Ningyu Zhang drafted §1 and §7, Yunzhi Yao drafted §2, §3 and §6, Yunzhi Yao and Zekun Xi drafted §4 and §5. Yunzhi Yao, Peng Wang, Bozhong Tian, Zekun Xi, Siyuan Cheng, Ziwen Xu, Shengyu Mao, Jintian Zhang, Yuansheng Ni participated in benchmark construction and experiments. Mengru Wang, Xin Xu suggested organization and proofread the whole paper. Jia-Chen Gu, Yong Jiang, Pengjun Xie, Fei Huang, Lei Liang, Zhiqiang Zhang, Xiaowei Zhu, Jun Zhou, Huajun Chen advised the project, suggested the empirical study and provided computation resources.

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